



OPO STARTUPS · JULY 2026

# AI-Powered Hybrid Search & Agentic Assistants

*on Cloudflare's Edge*

Build production search and durable AI agents that scale to zero —  
and back up.

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Senior DevOps Engineer · Applied Systems

# Why I Built This — It Starts at the Workbench

At **Juiced Wheels** a bench tech has a customer's e-bike on the stand and the clock running — and the answer is buried in a parts catalog and a stack of manuals no keyword box can navigate.

## 84K<sub>+</sub>

### parts to search

The right innertube is one SKU among tens of thousands across QBP · JBI · OSCO.

## 2,845

### PDF manuals

The fix is on page 47 of a Bafang PDF — not in a tidy, queryable database.

## "30h?"

### cryptic error codes

Bafang, Bosch & Shimano each speak a different dialect — the code is meaningless without the table.

## ≠

### symptom vs. keyword

"Motor cuts out climbing" isn't a phrase in any manual — it's a trace through the system.

A bench tech doesn't want ten documents — they want **the answer, cited, in seconds**, and bench time is a shop's biggest cost. That's the bar generic search couldn't clear, so I built **Shop Shepherd** on Cloudflare for ~\$80/mo — and the stack underneath **isn't bike-specific**.

# ! The Problem with Traditional Search Infrastructure

⚠️ EVEN WHILE YOU READ THIS SLIDE...

## \$0.0000

Burned by always-on infra  
(*\$1,950/mo baseline, \$0.0007/sec*)

## 0

Queries actually served

## 100%

Of spend on idle capacity



24/7

### Elasticsearch / OpenSearch

Dedicated clusters running 24/7 even with zero traffic.

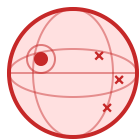
**\$200–800+/mo minimum, no queries needed**



### Pinecone / Weaviate

Vector DB subscription bills for **all 30 days**, used or not.

**\$70–250+/mo serverless tier**



1 REGION

### Regional Lock-in

Pinecone, ES clusters pin to one region. EU/Asia queries pay **200ms+** tax.

**Multi-region · 2–3× the cost or self-managed replication**



2 SYSTEMS  
custom merge code

### Stitched Stack

Vector DB + keyword search are **separate systems**. Four vendors, four bills, glue code to merge.

**No native hybrid · sync drift · double the ops**

# CF Why Cloudflare? The Edge Advantage

✓ SAME SLIDE, ON CLOUDFLARE'S EDGE...

**\$0.0000** —

Always-on cost

(flatline — pay only for what runs)

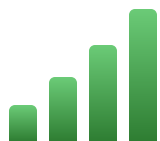
**1,247,832**

Queries served while you read

**\$24.96**

Total spend

(~\$0.00002 / query)



## Usage-Based Pricing

Pay per request, per query, per GB stored — not per hour.

Zero traffic = zero cost



## Global Edge Network

Workers run within ms of the user, not in us-east-1.

330+ PoPs worldwide



## Unified Platform

Workers + D1 + Vectorize + KV + R2 in one ecosystem.

One bill, no glue code



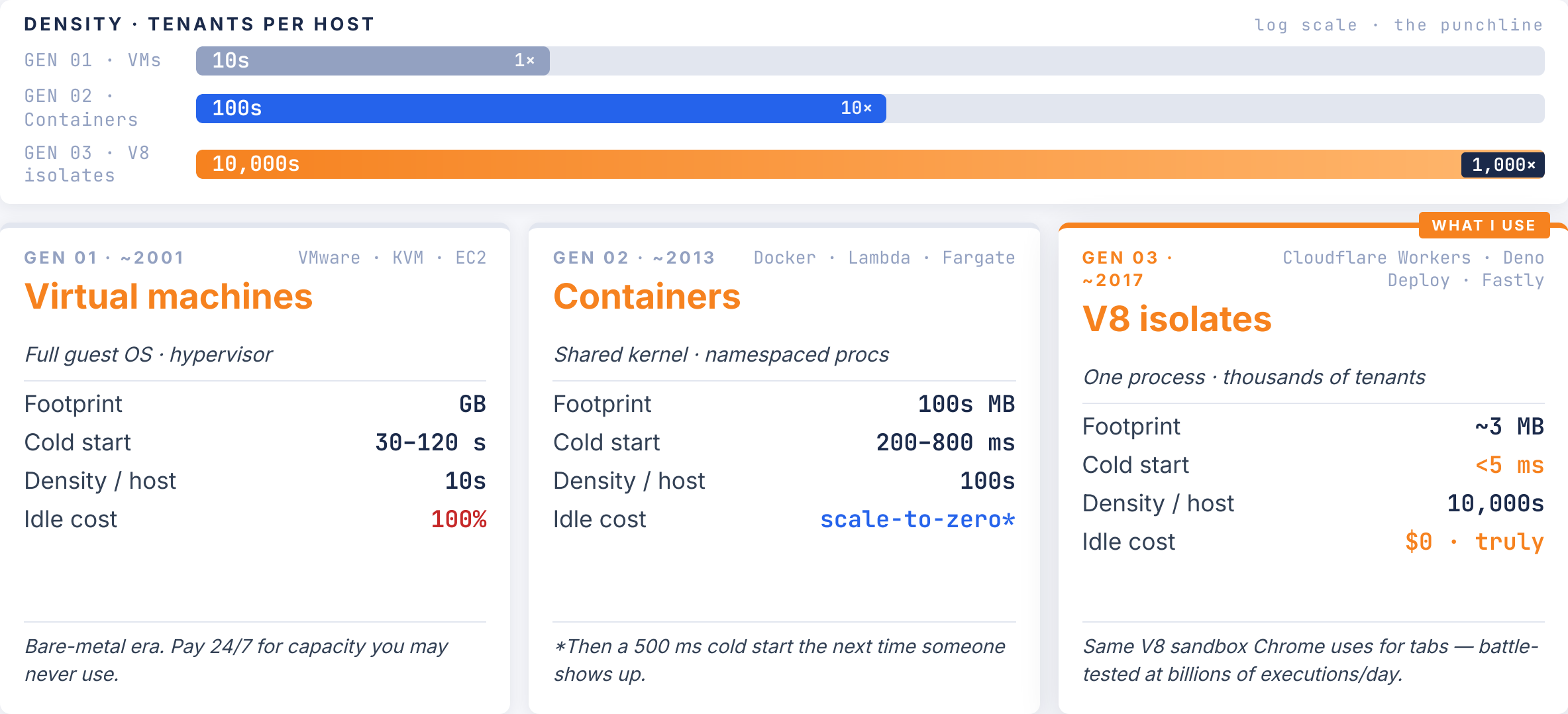
## Zero Cold Starts

V8 isolates start in sub-ms — the shop lights flick on, no warming a cold engine.

~0ms cold start vs ~200ms containers

# Three Generations of Compute

Each one cut cost and blast radius by an order of magnitude. V8 isolates are what makes scale-to-zero affordable at edge density.



PART ONE

# Search Architecture Deep Dive

*Three engines, one router — all on the edge*

# \$ The Search Stack — Think of It Like Shopping

Three search engines, three shopping behaviors. The customer's voice picks the station: semantic for symptoms, faceted for browsing with constraints, keyword for exact parts.



Measured engine times, warm queries. <sup>†</sup>Semantic embed: ~3ms cached / ~200ms cold. <sup>‡</sup>Meili is remote-hosted (NYC); warm round-trip adds ~400-700ms — FTS5 + Vectorize are co-located in the worker.

# ~ Layer 1: Find the Right Aisle — Vectorize

LAYER 1 OF 3

## Why Vector Search Matters

Keywords fail when users describe problems instead of naming parts. "My bike has a flat tire" should match "innertube replacement" — vector search makes this connection.

Workers AI maps each sentence to **768 numbers** — coordinates on a "meaning map," where similar meanings land near each other even with zero shared words. **embeddinggemma-300m** generates them; Vectorize finds the nearest via cosine similarity.

## Embedding Pipeline

- 1 Auto-chunk content > 7,500 chars
- 2 Workers AI → 768-dim vector
- 3 Vectorize index → stored at edge
- 4 KV cache → skip re-embedding

VECTORIZE STAGES

1 embed (Workers AI)

→

2 cosine search

→

3 topK=10

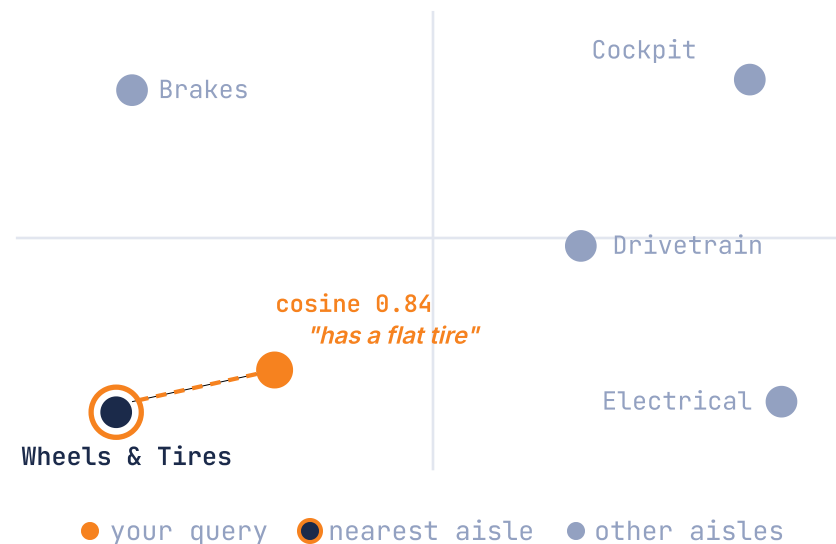
## Semantic Routing

CUSTOMER ASKS

*"My bike has a flat tire"*



MEANING MAP • NEAREST AISLE WINS







# Layer 2: Pick Off the Shelf — D1 + Meilisearch

## What Faceting Is

Filtering a big catalog by structured attributes — brand, wheel size, valve type, price, in-stock — with a **live count on every option**. It's how a tech narrows 84,000 parts down to the 8 that fit, without typing a perfect query.

## Why D1 *and* Meilisearch?

### D1 system of record

Normalizes 84K parts from 3 vendor feeds, scores e-bike relevance, and serves keyword search via FTS5 — the fallback if Meili is ever down.

### Meilisearch the speed layer

A sub-30ms count per facet checkbox across all 84K rows — more than a D1 `GROUP BY` can handle — plus typo tolerance ("shrader" finds "Schrader"). It's a faceted-search engine with its own hybrid keyword + vector ranking.

## Faceted Shelf in Action

CUSTOMER ASKS

*"Innertube — 26" × 4", Schrader valve"*

### FILTERS APPLIED

|                                                          |    |
|----------------------------------------------------------|----|
| <input checked="" type="checkbox"/> Tire width: 4.0"     | 84 |
| <input checked="" type="checkbox"/> Wheel diameter: 26"  | 46 |
| <input checked="" type="checkbox"/> Valve type: Schrader | 24 |
| <input checked="" type="checkbox"/> In stock             | 8  |

 **26" × 4.0–4.8" Fat Bike Tube**  
Q-Tubes TU1200 · Schrader valve · \$18.99

*typo-tolerant · ~28ms engine · fallback to D1 FTS5*

### THE LONE THIRD-PARTY

**Meilisearch (\$30/mo)** is the only non-Cloudflare piece of this whole stack — today's one platform gap for fast faceting + typo tolerance. If Cloudflare ships a native equivalent, this line drops to **\$0** and the architecture is 100% first-party.

MEILISEARCH STAGES

1 parse facets



2 filter (typo-tolerant)



3 rank by relevance

# Layer 3: Scan the Barcode — D1 + FTS5

LAYER 3 OF 3

## How It Works

D1 is Cloudflare's serverless SQLite; FTS5 is its built-in full-text search — co-located with your Worker, no extra service, no extra bill.

### FINDING · PROGRESSIVE RELAXATION

one round trip

|        |                     |
|--------|---------------------|
| EXACT  | "TU1200" → 1 hit    |
| PREFIX | "TU120..." → 6 hits |
| FUZZY  | "TU~1200" → 18 hits |

### RANKING · HOW BM25 SCORES A HIT

3 signals → one score

|           |                                        |
|-----------|----------------------------------------|
| RARITY    | "TU1200" outweighs a common "tube"     |
| FREQUENCY | more hits help — but less each time    |
| BREVITY   | a title hit beats one deep in a manual |

## Progressive Lookup

CUSTOMER ASKS

*"I need part TU1200"*



**TU1200**

Fat Bike Tube · 26×4 Schrader

*BM25 score 8.4 · 7 ms (measured)*

FTS5 STAGES

1 tokenize



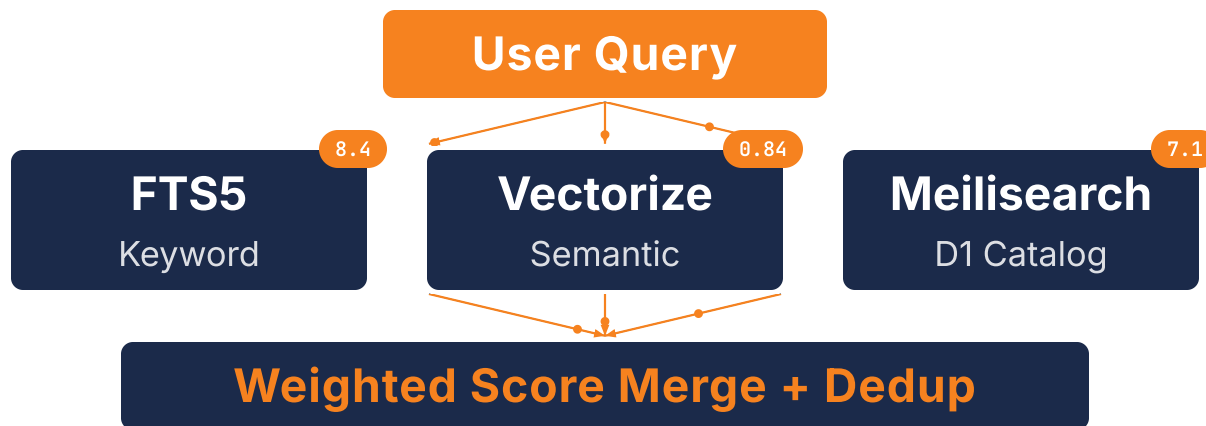
2 match (virtual table)



3 rank (BM25)

# ⊕ Hybrid Search: Merging Keyword + Vector

When one engine isn't enough, fire all three in parallel and merge by weighted relevance score. The agent never sees three result lists — just one ranked output.



TOP RESULT | **TU1200** Fat Bike Tube — matched by all three engines

final 18.2

## Source Weights

Manuals 2.0× · Docs 1.5× · Parts 0.2×

- OEM manuals win every tie — they're the authoritative source for technicians
- Parts get demoted unless the query is clearly a SKU lookup
- A result returned by multiple engines stacks the boosts

## Short Query Optimization

1-2 words

70% keyword

30% vector

3+ words

50% keyword

50% vector

- Short queries lean keyword — prevents false matches like *"brake"* ≠ *"break"*
- Below-floor results get cut, with a top-3 fallback so the agent never starves

# PDF Pipeline: OCR, Extract, Optimize

## Why OCR Every PDF?

Most OEM service manuals are scanned images or flattened exports — there's no selectable text layer. **OCR (Optical Character Recognition)** turns those page images into searchable text. I then inject that text **back into the PDF as a hidden layer**, so the manual is fully searchable in any viewer while looking identical to the original. **You can't chunk what you can't read.**

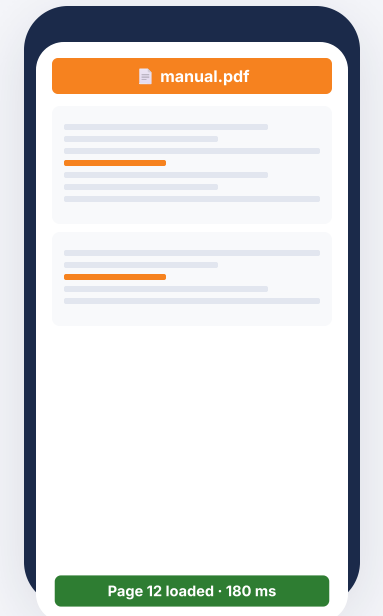
### RAW OEM PDF → MOBILE-READY



40× smaller · same content · ready for a 4G phone at the bench

### HOW I GET THERE

- ✓ Subset embedded fonts (*strip unused glyphs*)
- ✓ Recompress images (*JPEG2000 / lossy DCT*)
- ✓ Inject OCR text layer (*hidden, selectable, indexable*)
- ✓ Optimize to PDF/A (*archival standard — keeps the OCR text map*)
- ✓ Web-linearize (*first page renders before full download*)



*Tech reads at the bench*

PDF STAGES

1 OCR



2 chunk + embed



3 D1 + Vectorize



4 R2 (mobile + thumbs)



# Why Chunking Context Matters

## The Naive Chunking Problem

Most tutorials split documents into fixed-size blocks and embed them as-is — like cutting a manual into unlabeled index cards. You get a match back, but no idea which card it's from.

An orphaned chunk like *"torque the bolts to 40 Nm"* is useless without knowing it's a Bosch CX mid-drive install vs. a Shimano hub motor teardown.



Long manual PDF · 142 pages

Bosch CX Mid-Drive Install Guide

### ↓ chunked + tagged

chunk\_0042

"Torque bolts to 40 Nm. Check alignment..."

bosch-cx install

chunk\_0043

"Speed sensor mounting: 5 Nm. Use anti-seize..."

bosch-cx sensor

chunk\_0044

"Battery seat alignment: shim to 0.2 mm..."

bosch-cx battery

## Naive Chunk (no context)

```
{
  text: "Torque bolts to 40 Nm.
        Check alignment before
        tightening...",
  vector: [0.23, -0.41, ...]
}
```

## Metadata-Tagged Chunk (structural context)

```
{
  text: "Torque bolts to 40 Nm...",
  vector: [0.23, -0.41, ...],
  metadata: {
    title: "Bosch CX Installation",
    path: "/guides/bosch-cx-install",
    tags: ["mid-drive", "torque"],
    category: "installation",
    source: "manual"
  }
}
```

NEXT

Metadata fixes naming. But what about OCR garbage where even the chunk text is garbled? →

# Contextual Retrieval: Quality Up, Cost Down



Metadata gives chunks a structural home. A 1–2 sentence **LLM summary** — Anthropic's Contextual Retrieval pattern — adds a sticky note that stays findable even when the page is OCR garbage.

**Labor division:** *LLMs author chunk context; humans author the graph (slides 19–21) — enrich text, don't invent ontology.*

## BEFORE — RAW OCR CHUNK

### Indexes can't find it

```
"340 % 8 1 240% TURBO 340%
3 4 0 SPEED SENSOR BDU
4xx INSTALL torque 5Nm..."
```

- ✗ **FTS5 BM25:** 0 useful matches — query terms don't appear in garbled OCR
- ✗ **Vectorize:** embedding of garbage = garbage similarity

*Query: "Bosch Drive Unit speed sensor installation"*

## AFTER — CHUNK + context\_llm

### Both indexes hit at rank 1

```
context_llm
Bosch DriveUnit BDU4xx (MY24) installation guide — speed sensor mounting and
torque spec.
```

```
"340 % 8 1 240% TURBO 340%..."
```

- ✓ **FTS5 BM25:** "Bosch", "DriveUnit", "speed sensor" all match the summary
- ✓ **Vectorize:** coherent summary embeds cleanly → high cosine sim

## RETRIEVAL QUALITY · n=30 cases

### Top-1 accuracy

66.7% → **93.3%** +26.6 pp

### Top-3 accuracy

83.3% → **100%** +16.7 pp

## COST PER TURN · fewer tokens reach Sonnet

### Dollars per turn

\$0.201 → **\$0.075** -63%

### Sonnet synthesis tokens

49.6K → **13K** -73%

## PART TWO

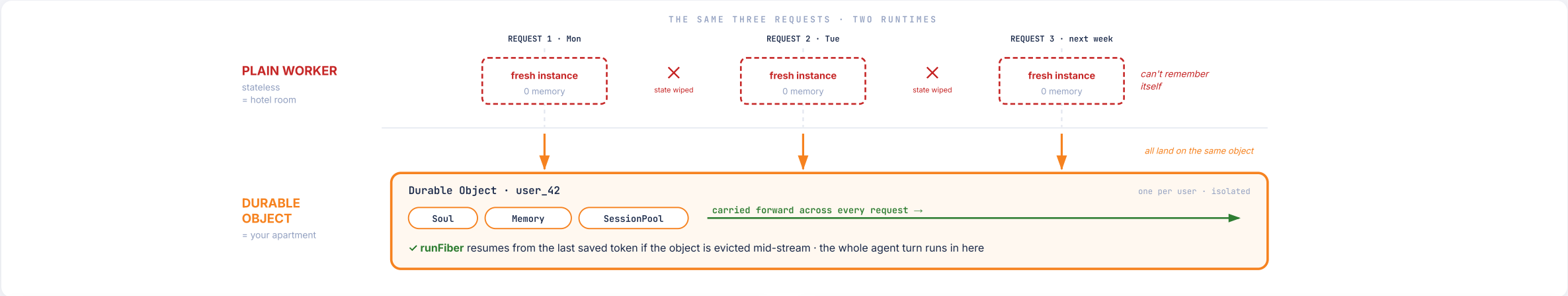
# From RAG to Agentic AI

*Everything so far was stateless — every request started cold. Now I add what search can't have: durable per-user memory, crash recovery, and query-aware context compression. Watch one number fall the whole way down:*

**~36¢ → ~3¢** per tool turn

# A Durable Object Per User (Project Think)

A Worker is a **hotel room** — fresh every request, your stuff is gone. A Durable Object is your **apartment** — same address, same key, your memory is still on the counter.



| TOOL                 | PURPOSE                         |
|----------------------|---------------------------------|
| search_documentation | unified hybrid search           |
| search_manuals       | PDF manual sections             |
| search_parts         | faceted parts catalog           |
| diagnose             | symptom → ranked causes (graph) |
| read_manual          | deep-read one manual            |
| get_content_section  | fetch a known section           |
| search_internet      | web fallback — recalls, news    |

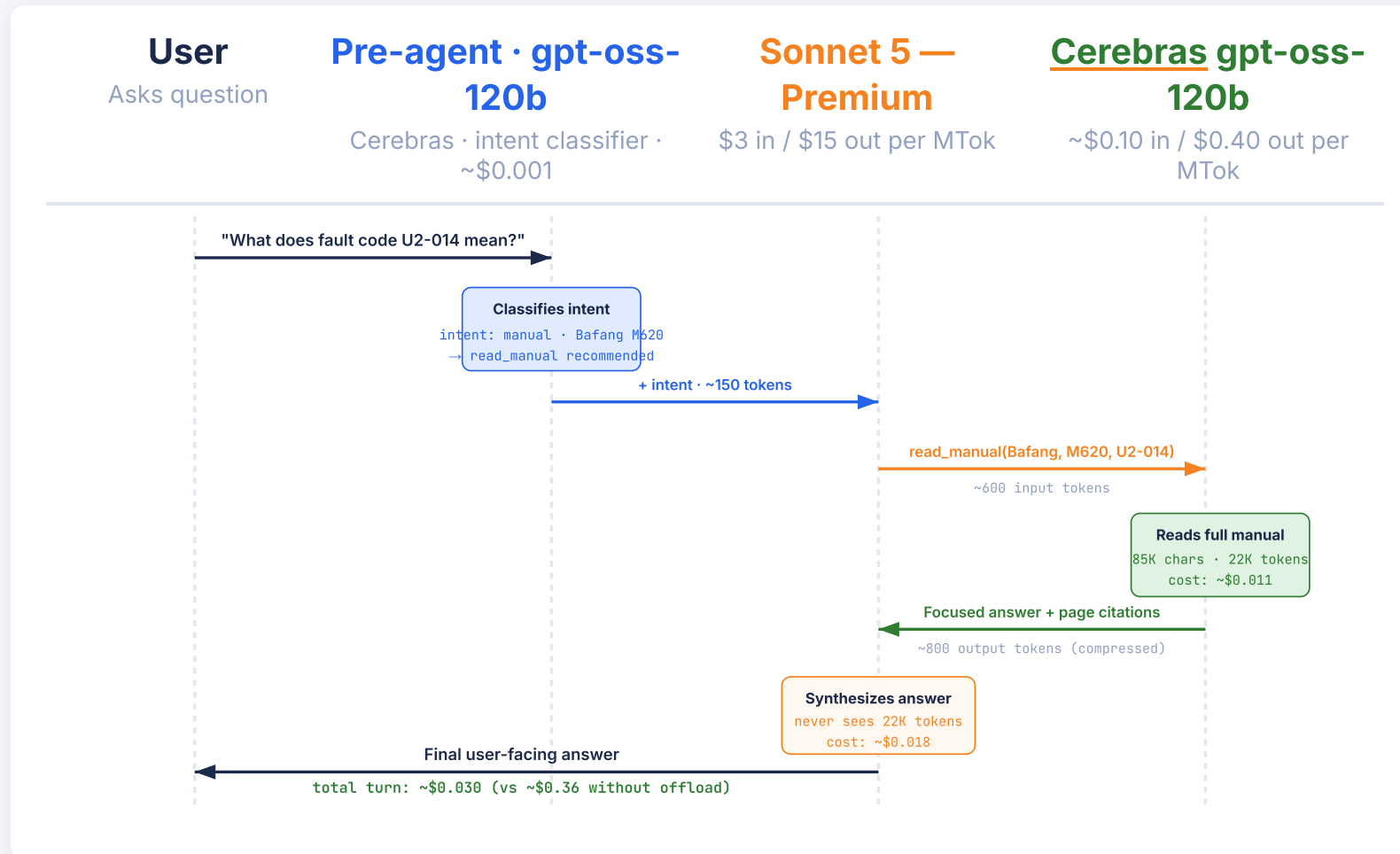
| PER-USER DURABLE OBJECT STATE                                                   |
|---------------------------------------------------------------------------------|
| <b>Soul block</b><br>System prompt from KV — live-refresh per turn              |
| <b>Memory block</b><br>Per-user facts via UserMemoryProvider — durable identity |
| <b>SessionPool</b><br>Per-thread history, FTS5-searchable                       |
| <b>chatRecovery</b><br>runFiber — mid-crash saves partial text                  |



# ↔ Sub-Agent Offloading: Cheap Bulk Work, Premium Synthesis

High-token tool work goes to a near-free model; the premium model only sees the compressed result and writes the answer. Like a shop, the apprentice does the bulk grunt work and the master tech makes the call.

~\$0.36 → ~\$0.030 per turn — a 90% cut, same answer quality.



## THROUGHPUT TOKENS / SECOND

Cerebras gpt-oss-120b (used here)

~3,000 t/s

20–60× faster

Groq · Together (specialized inference)

~500–1,000 t/s

~7×

Sonnet 5 · GPT-5 · Gemini 2.5 (frontier APIs)

~50–150 t/s

1× baseline

20–60× faster than frontier APIs.

The pre-agent adds ~250ms, not ~3 seconds — that's what makes the offload pattern viable.

**Why Cerebras for the mini-agent:** I trialed Claude Haiku and Cloudflare Workers AI models too — both viable, but Cerebras's speed at low cost was the best fit for work that runs on every turn.

# ↑ Query-Aware Compression: 90% Cheaper, 10× Sharper

## UNIT ECONOMICS

### × BEFORE · DUMB TRUNCATION

#### The model gets noise.

*Like clipping the manual at page 16.*

- Tool returns **15 KB** raw PDF text
- Char-count clip at 16K characters
- Sonnet sees 8 irrelevant pages of OEM filler
- Burns **8 follow-up tool calls** narrowing down
- No page citations, no section anchors

Response: 230 chars of vague advice

### ✓ AFTER · QUERY-AWARE COMPRESSION

#### The model gets answers.

*Like a service writer marks the right page.*

- Same 15 KB result — pre-Sonnet hop adds `gpt-oss-120b`
- Cerebras sub-agent extracts query-relevant sections
- **Page citations + chunk IDs preserved**
- Sonnet receives **3 KB** of focused content
- Answers in **2 calls** with diagnostic depth

Response: 2,239 chars of specific guidance

#### INPUT TOKENS

~~15K~~ → **3K**

per tool result sent to Sonnet

-80%

#### TOOL CALLS / ANSWER

~~8~~ → **2**

no follow-up narrowing

-75%

#### COST / TURN

~~\$0.36~~ → **\$0.03**

≈ \$330 saved / 1K turns

-90%

#### USEFUL RESPONSE

~~230~~ → **2,239**

chars of specific guidance

+10×

**RAG retrieves. Prompting reasons. Compression curates.**

Always-compress tools: `read_manual` · `get_content_section` · others compress only above the 4K-token budget

# When retrieval *isn't* enough.

Search returns **text**. Diagnosis needs **causality** — what supplies what, what cuts what, what symptom traces to which subsystem.

## 01 WHAT I HAD

Hybrid search over 2,845 manuals, 2,334 articles, 84K parts.

- FTS5 + Vectorize + Meilisearch fused via `/api/search/unified`
- Weighted merge + **Gemini structural rerank** — ranked passages with page citations

Great at finding the **page**. Useless at finding the **cause**.

## 02 WHAT BROKE

"Motor cuts out climbing" returned 12 pages.

- Pages described the **symptom** (motor cutting out under load)
- None named the **cause** (controller overheating) or the **subsystem** (thermal cutoff disabling motor power)

A symptom isn't a keyword. It's a **trace** through the system.

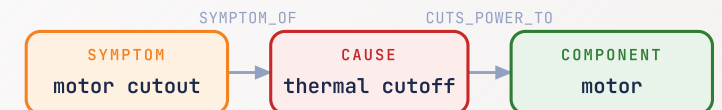
## 03 WHAT I BUILT

A diagnostic graph built on physics, real-world conditions & rider behavior.

- 825 profiled nodes — *each carries tools · symptoms · tiered \$0-first steps*
- 2,523 typed edges — causal verbs the agent calls per turn via `query_ebike_graph`

SUPPLIES\_POWER\_TO · CUTS\_POWER\_TO ·  
SYMPTOM\_OF · SENDS\_SIGNAL\_TO · TESTED\_BY

ONE CAUSAL TRACE, WALKED PER QUERY



## WHY HAND-CRAFTED

I tried LLM extraction. It hallucinated. I shipped hand-crafted instead — and the agent still queries the graph every diagnostic turn. Here's the trade-off, the anatomy, and the live trace.

# Why *hand-crafted* beat LLM extraction.

## LLM EXTRACTION · TRIED IT

### High cost. High risk.

DEAD-END

- **Hallucinated edges** (`motor PART_OF tire?`) → fabricated repair advice
- 963 docs → `Anthropic Batch API` → entities + relationships
- 31 duplicate "Battery" variants → canonicalization nightmare
- No auto-sync — graph drifts every time a doc changes
- ~\$40 per full re-extraction × every doc update
- No audit trail — can't diff a node's history

Kept it OFF in production



## HAND-CRAFTED · SHIPPED IT

### \$0 cost. 100% deterministic.

SHIPPED

- 825 nodes in TypeScript — one file, one source of truth
- Expert-authored with e-bike diagnostic ordering
- Regenerate SQL + seed in seconds — zero LLM cost
- Zero hallucinated edges → no fabricated repair advice
- Every change is a PR with a reviewable diff
- Agent still uses GraphRAG to `traverse` — humans author, LLMs query

LIVE in production, every diagnostic query



” Truth is human work. Reasoning over it is LLM work. ”

## INTEGRATED WITH EVERY DIAGNOSTIC QUERY

```
diagnose('motor cutout climbing') → sym_motor_cutout_under_load (12 ms · D1)
```

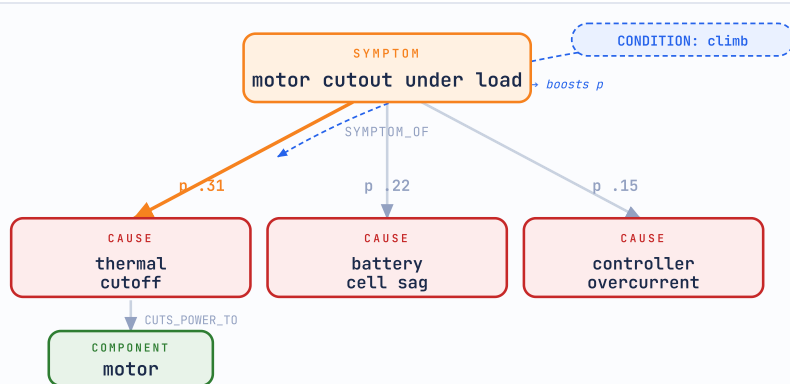
```
query_ebike_graph('controller', 'interrupts') → brake_cutoff_sensor, shift_sensor (safety overrides)
```

```
lookup_error_code('E021') → controller_thermal_cutoff (69 codes indexed)
```

# Not a search index — a **diagnostic brain**.

825 expert-authored diagnostic profiles · 2,523 weighted causal edges — *error codes are just one way in.*

## ONE DIAGNOSTIC SLICE · D1 + KV



Think of it as the bike's wiring diagram the AI can walk: parts are the **nodes**, the labeled wires between them (**typed, weighted edges** — SYMPTOM\_OF · CUTS\_POWER\_TO) are the links, and the agent *traces the circuit* from symptom to cause instead of reading a paragraph that happens to mention the part. A real slice of the 825-node graph; live **conditions** (a climb) re-weight the paths. **Expert-authored, not LLM-scraped.**

symptom cause / failure component edge = relation · p = probability

### ALSO IN THE GRAPH

18 node types · 85 typed relations · environmental multipliers (rain · cold · cargo)

SYMPTOM\_OF · CAUSES\_FAULT\_IN · FAILS\_AS · TESTED\_BY · INTERRUPTS · CUTS\_POWER\_TO

## A PLAYBOOK PER NODE — thermal\_cutoff

|             |                                                  |
|-------------|--------------------------------------------------|
| TOOLS       | multimeter · IR thermometer · torque wrench      |
| CONSUMABLES | dielectric grease · thermal paste                |
| SYMPTOMS    | cutout under load · power returns after cooldown |

|                |                                                 |
|----------------|-------------------------------------------------|
| TIER 0 · \$0   | temp check after a climb · airflow + brake drag |
| TIER 1 · meter | temp-sensor resistance · connector continuity   |
| TIER 2 · swap  | replace thermal sensor / controller             |

Every node carries this — cost-tiered, e-bike-specific, cheapest-first.

## HOW GRAPHRAG ANSWERS — "motor cuts out climbing"

- MATCH** **diagnose** maps the phrase to a **node** — codes & the 768D vector are other ways in → `sym_motor_cutout_under_load`
- TRAVERSE** **query\_ebike\_graph** walks causal edges to pull the connected causal subgraph
- RANK** scored by base p × live conditions (climb = load + heat) → `thermal_cutoff · battery_cell_sag · controller_overcurrent`
- ANSWER** Sonnet reasons over the **causal subgraph** and names the cause + the fix

Plain RAG retrieves *passages*.

GraphRAG retrieves the **causal chain**.

# From symptom to *diagram* — live.

Shop Shepherd · shepherd.shopdawg.app

claude-sonnet-5

Customer says the motor cuts out when climbing hills. Bafang M620.

→ tool: diagnose

```
{ "query": "motor cutout climbing", "limit": 3 }
```

← sym\_motor\_cutout\_under\_load 12 ms · D1

→ tool: query\_ebike\_graph

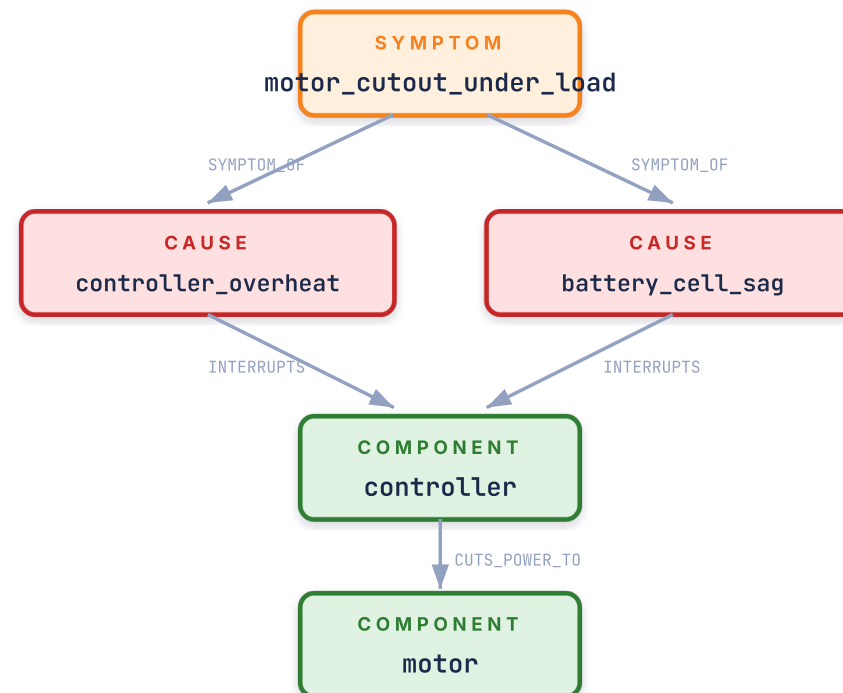
```
{ "node": "controller", "verb": "interrupts" }
```

← 4 nodes, 5 edges · rendering trace... 8 ms · D1

Likely **controller overheating** or **battery cell sag** — under sustained climbing load, pack voltage drops or the controller overheats, and its cutoff trips. Diagram on the right.

GENERATED MERMAID · RENDERED INLINE

graph TD



■ symptom

■ cause

■ component

→ LIVE DEMO

IF YOU DON'T WORK ON BIKES **motor** = the web server · **controller overheating** = CPU throttling under load ·

**battery sag** = DB / resource latency under load

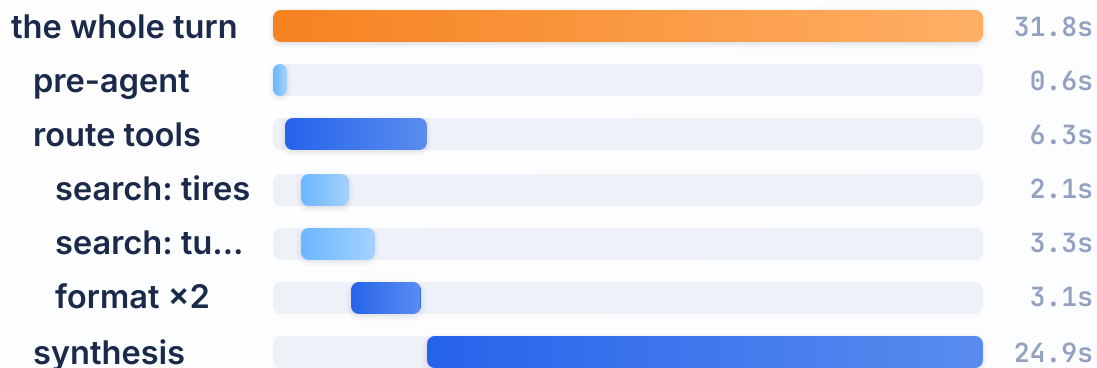
# Observability: One Pane of Glass

I was already running PostHog for product analytics — funnels, retention, session replay. Turning on its **LLM observability** dropped every AI trace onto that same dashboard: cost, tokens, latency, and tool-routing, per turn.

## ONE REAL TRACE · POSTHOG

"Best margin on 26×4 fat tires + tubes, grouped by distributor"

■ \$ai\_trace ■ \$ai\_generation ■ \$ai\_span



44.9K tokens · \$0.072 · 8 spans. The Sonnet synthesis alone is 24.9s — 78% of the turn (streaming a long answer, not a stall); the 3 searches + 2 Cerebras formatters total ~3s and \$0.009. You only see that once it's traced.

## ONE PANE OF GLASS

**AI traces land right next to the product analytics I already had.**

**Per-turn cost & tokens** — cache-aware, so prompt-cache savings show up

**Tool routing** — which tools fired, in what order, how long each took

**Latency per step** — pre-agent · tools · synthesis

Beside **funnels, retention & session replay** — already running

Recommended, not required — and **not part of the ~\$80/mo search infra**. PostHog was already in the stack; the LLM layer was icing on the cake.

[posthog.com/docs/ai-observability](https://posthog.com/docs/ai-observability)



# Real-World Scale: What This Handles

2,845

## PDF Manuals Indexed

Gathered with AI search — Brave · Tavily · Firecrawl  
OCR'd · ~593K chunks · full-text + vector indexed



2,334

## Repair Articles

Built from Google Deep Research, human-curated  
full-text + vector indexed



84K+

## Parts in Catalog

Nightly refresh from 3 vendor feeds  
D1 faceted, Meili-accelerated



\$80/mo

## Entire production stack

Workers + D1 + Vectorize + Meili included





\$

# Cost Comparison: Traditional vs. Cloudflare

UNIT ECONOMICS

WHERE THE MONTHLY SPEND GOES · REPRESENTATIVE PRODUCTION SCALE · JUN 2026 RATES



CLOUDFLARE'S \$80, ZOOMED IN 25× — WHERE IT ACTUALLY GOES



Orange = Meilisearch, the only paid third-party — skip it and D1 FTS5 covers parts (Workers Paid \$5 · D1 free tier). Infra floor only — LLM inference (Sonnet + Cerebras) adds ~\$50–200/mo at usage.

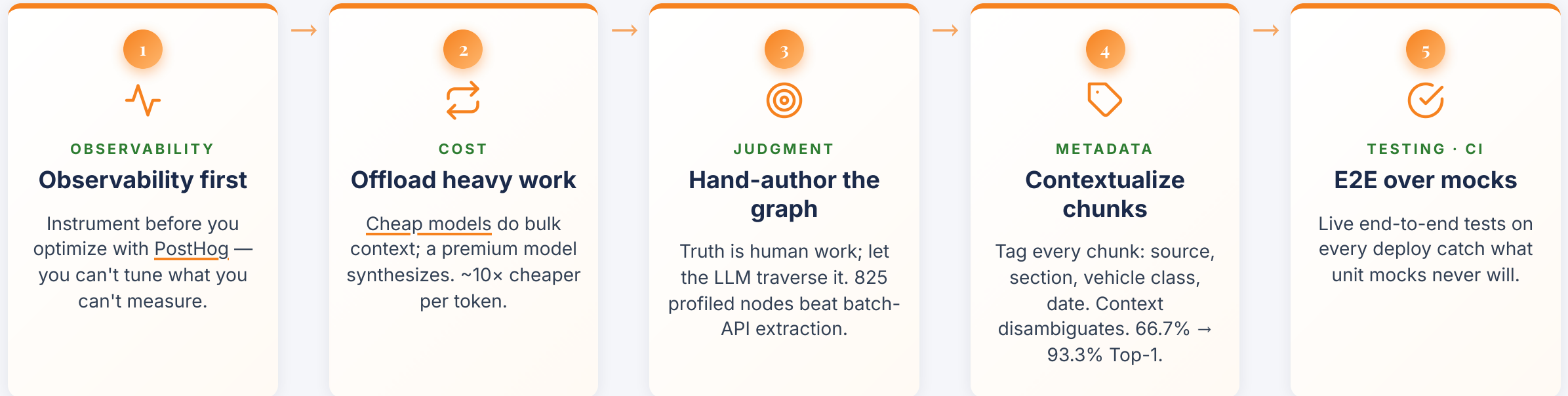
25× cheaper · same throughput · scale-to-zero baseline





# Lessons from Production

Five hard-won lessons from running this in production — **not the order I learned them, but what I'd hand my past self on day one.**



## THE ONE-LINE VERSION

Let **cheap models do the volume**, keep **humans on the judgment calls**, and **measure everything** in between.

PART THREE

# Build It Yourself

*Two paths, same destination — managed AI Search to start, raw primitives when you want full control.*

*None of it is bike-specific — the architecture holds for any domain.*

# Cloudflare *AI Search*.

The retrieval primitives you just saw — provisioned for you, fully managed.

## WHERE AI SEARCH FITS IN YOUR STACK

~80% of the retrieval pipeline without writing a Worker. It provisions the whole **retrieval pipeline** you just built:

✓ AI SEARCH

**Hybrid RAG — vector + BM25 keyword, reranked**

R2 · Workers AI · Vectorize · AI Gateway · query rewriting · cited answers

**Faceted parts**    Meilisearch · typo-tolerant filters

**Agent + Memory**    Durable Objects · diagnostic graph

↑ bolt these on in Step 2, as you outgrow the shortcut

Open beta · free within limits · 4 MB max per file

## IF YOU REMEMBER ONE THING

**Same R2 + Workers AI + Vectorize + AI Gateway pipeline.**

Click to provision. No wrangler glue.

Drop docs into a bucket. Auto-chunked, auto-reindexed.

Hit one endpoint. Ranked results or full RAG answer.

Same scale-to-zero economics. Same edge latency.

[developers.cloudflare.com/ai-search](https://developers.cloudflare.com/ai-search)

# Same primitives, *full control*.

Started on **AI Search**? Steps 1–4 (the whole hybrid retrieval pipeline) come provisioned — your real build is the **agent + memory + domain graph**.

1

DATABASE · SQL

**D1 + FTS5 for keyword** ✓ via AI Search

```
wrangler d1 create my-search-db
```

```
CREATE VIRTUAL TABLE ... USING fts5()
```

2

VECTORS

**Vectorize for semantic** ✓ via AI Search

```
wrangler vectorize create my-index \
```

```
--dimensions=768 --metric=cosine
```

3

INGEST · COMPUTE

**Worker to ingest & embed** ✓ via AI Search

Chunk with metadata, embed via Workers AI, write to D1 + Vectorize in one transaction.

4

SEARCH · API

**Hybrid endpoint** ✓ via AI Search

Parallel FTS5 + Vectorize, weighted merge by source, threshold + dedup.

5

AGENT · MEMORY

★ YOUR REAL BUILD

**Agent layer for tools & memory**

Install [@cloudflare/think](#) — streaming, tools, per-user Durable Object memory, crash recovery.

## GRADUATE TO STEP 2 WHEN ANY OF THESE IS TRUE

**Big docs** — PDFs past the beta's ~4 MB file ceiling

**Per-user state** — durable memory + crash recovery (Durable Objects)

**Custom scoring** — hand-tuned hybrid ranking / your own rerankers

**Causal diagnosis** — typed-edge GraphRAG, not semantic proximity

**Faceted catalog** — typo-tolerant facet filters over 84K parts (Meilisearch)

**Tool breadth** — a long tail of bespoke agent tools



# Start Building Today

Everything shown here runs on Cloudflare's free or pay-as-you-go tiers

Start on AI Search → grow into the primitives as you scale. Same edge, same economics, no rip-and-replace.

[developers.cloudflare.com/d1](https://developers.cloudflare.com/d1)

[developers.cloudflare.com/vectorize](https://developers.cloudflare.com/vectorize)

[developers.cloudflare.com/workers-ai](https://developers.cloudflare.com/workers-ai)

[developers.cloudflare.com/durable-objects](https://developers.cloudflare.com/durable-objects)

[blog: AI Search primitive](#)

[blog: Project Think](#)



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THESE SLIDES

See it live → [shepherd.shopdawg.app](https://shepherd.shopdawg.app)

## Built for e-bikes. The primitives fit any industry.

Same engine — healthcare is next.